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Study shows heat-related deaths rose in Florida nursing homes after Hurricane Irma

WUSF | By Leda Alvim

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Hurricane Irma impacted millions of people across Florida when it made landfall in 2017, leaving the state with power outages that lasted days, and in some places, weeks.

Power outages that followed Hurricane Irma affected 28,000 residents of Florida's nursing homes — and led to a 25% increase in deaths a week after the storm made landfall in September 2017.

Hurricane Irma impacted millions of people across Florida when it made landfall in 2017, leaving the state with power outages that lasted days and, in some places, weeks.

The outages also had a significant impact for hundreds of nursing homes across the state, which resulted in the death of some residents due to heat exposure.

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An article in the Nov. 24 issue of the *JAMA Health Forum* reported a 25% increase in nursing home deaths a week after Irma came on shore in Florida on Sept. 10, 2017, and a 10% increase in mortality rates 30 days after landfall.

Lindsay Peterson, an assistant professor at the University of South Florida School of Aging Studies, is a co-author of the study.

The study reported that about 28,000 nursing home residents age 65 and older were in facilities that lost power during Irma. The deaths that followed, according to Peterson, were mainly due to heat exposure caused when nursing homes lost air conditioning and couldn't power it with generators.

"The big concern in the past has been having enough power to run an oxygen machine, to run the things that are needed to help with life support — not so much having the power that's needed to run an air-conditioning system because that is huge," said Peterson..

"It takes a ton of power when you're in a big place ... to keep an air-conditioning system going. Now, some places did have that, [and] it was really interesting that some nursing homes were way ahead of the curve on this, [but] some were not."

Without air conditioning or other means of cooling, Peterson said, the temperatures inside nursing homes rose to dangerously high levels, which led to an increase of body temperatures in residents.

"Someone can become overheated in a matter of hours, and it sets in motion a series of processes that can harm them very quickly," Peterson said. "Older adults cannot regulate their body temperatures as well as everyone else. Once [body temperature] gets to a certain level, it's very, very difficult to bring that down."

The study specifically cited an incident at the Rehabilitation Center at Hollywood Hills, a nursing home near Miami, where 12 residents died in the days following Irma after the air conditioning failed. More than 150 other residents were evacuated from the facility because of the conditions inside, and the state eventually revoked the center's license to operate.

The state responded by passing a law mandating that long-term care facilities have backup generators to run air-conditioner units.

The Emergency Environmental Control for Assisted Living Facilities rule, according to Peterson, led many nursing homes to implement better emergency plans for the future.

"There is a much greater degree of attention to this and safety. If a hurricane were to hit [now] ... I don't believe we would see the same level of mortality," Peterson said.

The rule requires nursing homes and assisted living facilities to have the ability to keep

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outage to prevent residents from becoming overheated.

Facilities were required to have implemented their plans by no later than June 1, 2018 — but not all have done so. As of today, 99.7% of all assisted living facilities and 98.6% of nursing homes are in full compliance with state requirements.

While generators are part of the solution, Peterson said, the most important thing is to be able to get electrical power on again as soon as possible.

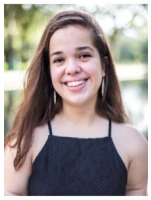
“Generators are not really the answer,” she said. “They help a great deal in the days after a storm, but if the power loss is extensive, you're going to need a lot of fuel, and that might be difficult. So there are continuing concerns around ensuring that there is enough power when a power loss is as massive as it was during Hurricane Irma.”

Peterson also hoped the study would increase awareness about these vulnerable populations and how similar events in the future could have deadly impacts if not addressed.

“We need a coordinated effort to take care of them in every way during a hurricane that involves every level of government and also people in the community who can come out to help if needed,” she said.

“We need to be aware that the storms that are coming are more powerful and we need to be prepared to deal with a situation where we have power loss on such a massive scale.”

Tags Health News Florida | Assisted Living



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